

IBM University Engagement Project Submission

Topic :- Fake News Detector for Students

Project Tile – Fake News Detector for Students

Domain of Project – Education

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Problem Statement -Fake News Detector for Students

The challenge

Volume & Speed: New content is created and shared every second, making manual fact-checking impossible for students.

Complexity: Fake news often mixes truth with falsehood, uses emotional language, or mimics real news sites, making it hard to detect.

Bias & Context: Articles can be misleading due to biased reporting, out-of-context images, or manipulated headlines.

Accessibility: Existing fact-checking tools are technical and not designed for students who need quick, simple, and reliable explanations.

The Objective

Build an AI-powered Fake News Detector that helps students:

Analyze uploaded or pasted news articles and social media posts for credibility using NLP and machine learning models.

Assess reliability by checking factors like source reputation, language patterns, factual consistency, and cross-referencing with trusted sources.

Summarize the content into a concise, easy-to-understand version that highlights key facts and flags suspicious claims.

Educate students by explaining why a piece of news is likely fake or reliable, promoting critical thinking and media literacy.

Technology Used

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- *AI/ML*: BERT, Hugging Face, LLMs for text analysis & summary
- *Backend*: Python, FastAPI, BeautifulSoup for scraping
- *APIs*: Google Fact Check API, NewsAPI for verification
- *Frontend*: <http://React.js>, Chrome Extension for easy student access
- *Techniques*: NLP, Sentiment Analysis, Source Reputation Check

Proposed Solution -

Build an AI web app + Chrome extension where students can paste a news link or text.

How it works:*

- *Input*: Student enters URL/text of news article
- *AI Analysis*: Checks language tone, source reputation, and cross-verifies facts with trusted sites
- *Result*: Gives Credibility Score: Real / Suspicious / Fake
- *Summary*: Shows short, simple summary + highlights red flags
 - *Learn*: Explains _why_ it's fake/real to improve student awareness

Key Benefit: Fast, simple, student-friendly tool to stop misinformation at the source

component Used -

1. **Input**: Text/URL box + scraper
2. **AI Engine**: NLP model + Fact-check API + Source checker
3. **Output**: Credibility score + Simple summary + Red flags
4. **Interface**: Web app + Chrome extension
5. **Database**: Stores queries & flagged news

workflow -

- ***Input*** → Student pastes news URL/text
- **Extract*** → Get article content + source
- ***Analyze*** → NLP + Source + Fact Check
- **Score*** → Mark as Real/Suspicious/Fake
- ***Display*** → Green/Yellow/Red + summary + red flags

Role in the solution:-

- Detects* → Auto-flags fake news in 3 sec
- Explains* → Shows _why_ with red flags + summary
- Prevents* → Stops sharing of misinformation
- Teaches* → Builds media literacy skills

Uniqueness of the solution:-

- Student-first → Built for teens, not journalists. Simple 3-line summaries + color results
- Explains, not just flags → Shows why it's fake with red flags and bias checks
- 3-second speed → Instant results before they hit "Share" on WhatsApp
- All-in-one check → NLP + Source + Fact verification in one tool
- Teaches while detecting* → Every result builds media literacy

Git Hub Link

<https://github.com/Pragya-200514/Fake-news-detector-for-students-./tree/main>

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Future Scope

- Multi-language* → Hindi, Marathi, other regional languages for wider reach
- Platform integration* → Direct WhatsApp/Instagram plugins to check before sharing
- Image/Video check* → Detect fake images, deepfakes, and doctored videos
- School dashboard* → Analytics for teachers to track trending fakes in class
- Voice input* → Check news by speaking or forwarding voice notes

Thank You!

Thank you for your time and interest.